

Junqiao Fan

Academic Curriculum Vitae

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CONTACT

Junqiao Fan
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CURRENT POSITION

PhD Student
School of Electrical and Electronic Engineering
Nanyang Technological University
Supervisor: Prof. Lihua Xie
Co-supervisor: Prof. Jianfei Yang

EDUCATION DATES

Ph.D., NTU
TODO–Expected TODO
B.Eng., The Hong Kong Polytechnic University
TODO–TODO

Last updated: June 7, 2026

RESEARCH OVERVIEW

My research focuses on human-centered AI, enabling AI agents, robots, IoT systems, and healthcare devices to perceive, understand, and interact with humans. My long-term vision is to build warm and trustworthy AI assistants with strong human awareness, supporting daily living and healthcare. My current work advances this goal along three axes: privacy-preserving motion capture using low-cost RF sensors such as mmWave radar; human motion modeling with advanced generative AI; and healthcare AI for monitoring physical and mental well-being with IoT devices.

RESEARCH INTERESTS

- Human-centered AI; multimodal human sensing; radar-based perception.
- Human motion analysis, human mesh reconstruction, and motion prediction.
- Generative models, diffusion models, dataset distillation, and healthcare AI.
- Human–robot interaction, embodied AI, IoT systems, and privacy-preserving sensing.

EDUCATION

Nanyang Technological University

TODO–Expected TODO

Ph.D. in Electrical and Electronic Engineering

School of Electrical and Electronic Engineering. Supervisor: Prof. Lihua Xie. Co-supervisor: Prof. Jianfei Yang.

The Hong Kong Polytechnic University

TODO–TODO

B.Eng. in Computer Science

TODO: add GPA, thesis title, or honors if needed.

KEY PUBLICATIONS

Federated Learning Driven Secure Internet of Medical Things. Junqiao Fan, TODO.
TODO venue / journal / year.

TODO: add DOI, arXiv, project page, or code link.

Diffusion Model is a Good Pose Estimator from 3D RF-Vision. Junqiao Fan, TODO.
TODO venue / year.

TODO: add acceptance information, DOI, arXiv, or project page.

mmPred: Radar-based Human Motion Prediction in the Dark. Junqiao Fan, TODO.
TODO venue / year.

TODO: add one-line contribution statement and link.

M4Human: A Large-Scale Multimodal mmWave Radar Benchmark for Human Mesh Reconstruction. Junqiao Fan, Yunjiao Zhou, Yizhuo Yang, Xinyuan Cui, Jiarui Zhang, Lihua Xie, Jianfei Yang, Chris Xiaoxuan Lu, Fangqiang Ding.
IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026.
arXiv: 2512.12378. TODO: add project page / code / dataset link.

Generative Dataset Distillation Using Min-Max Diffusion Model. Junqiao Fan, TODO.
TODO venue / year.

TODO: add DOI, arXiv, project page, or code link.

RESEARCH EXPERIENCE

Human-centered AI and RF Sensing Research

TODO–Present

Nanyang Technological University

Develop learning-based approaches for radar-based human sensing, motion modeling, and human-aware AI systems. TODO: add 2–3 concrete technical contributions.

Healthcare AI and Internet of Medical Things

TODO–Present

TODO Lab / Institution

Work on secure, privacy-preserving, and scalable IoMT systems for healthcare applications. TODO: add project details only if relevant.

SELECTED PROJECTS

- **M4Human:** large-scale multimodal mmWave radar benchmark for human mesh reconstruction.

- **mmPred:** radar-based human motion prediction under challenging sensing conditions.
- **RF-Vision Pose Estimation:** diffusion-based pose estimation from 3D RF observations.
- **Secure Internet of Medical Things:** federated learning for privacy-preserving medical IoT systems.

HONORS AND AWARDS

- TODO: add scholarship, best paper, competition, or travel award.
- TODO: remove this section if not needed.

ACADEMIC SERVICE

- TODO: reviewer for conferences / journals.
- TODO: teaching assistant / mentoring / workshop organization.

TECHNICAL SKILLS

Programming	Python, PyTorch, MATLAB, C/C++, ROS/ROS2, LaTeX.
Research Tools	mmWave radar processing, point clouds, SMPL/SMPL-X, diffusion models, multimodal learning.
Systems	Linux, Git, Docker, CUDA, distributed training.

REFERENCES

Available upon request.